Meridian Open University

Board Review: who enrols, how they engage, and where to focus in 2016

Executive summary

Only **47.2%** of 32,593 registrations end in Pass or Distinction. Withdrawal rose from **26.8%** (2013J) to **34.0%** (2014J); on the same modules the rise is to **31.4%**, because the weakest module, CCC, only started in 2014. The problem is not spread evenly: **50.9%** of withdrawals happen by day 30, and a third group we identified, the **silent stayers** (**10.2%**), never withdraw but go quiet before halfway, and **99.8%** of them fail. Meridian is effectively **two universities in one**: 60% of students stay engaged and 78% of them finish, while 40% disengage and only 1.3% finish. The 2016 focus should be the front door for students who quit, a mid-course re-engagement plan for silent stayers, workload at sign-up, and specific modules.

1. The business problem

Meridian is a distance-learning provider where students enrol in individual modules and interact almost entirely through a virtual learning environment (VLE), in which every click is logged. Roughly a third of registrations end in withdrawal, which has been treated as the price of the access mission. The Board asked: "Based on everything in this data, what kind of institution are we, and what should that tell us about where to focus in 2016?" The analysis answers this from the present data, not as a forecast or early-warning model.

2. The data

Table	Rows	What one row means	Key
studentInfo	32,593	A student in one module-run, with final result	module + presentation + student
studentRegistration	32,593	Registration and unregistration day	module + presentation + student
studentVle	10,655,280	Clicks by a student on one page on one day	none (rows summed)
vle	6,364	One VLE page and its activity type	id_site
studentAssessment	173,912	One submission and its score	id_assessment + student
assessments	206	One assessment: type, date, weight	id_assessment
courses	22	One module-run and its length in days	module + presentation

3. Methods used

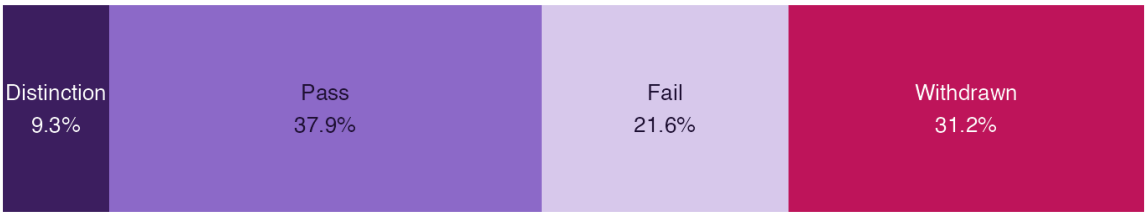
Step	Method	Purpose
Data checks	Primary-key uniqueness, 7 join (anti-join) checks, missing-value counts	Confirm the seven tables can be safely linked
Cleaning	Relabel "10-20" band, missing IMD as "Unknown", order categories, fill 11 blank exam dates with module length, never count banked results as late	Consistent, chart-ready data
Master table	Aggregate 10.6M click rows and 173,912 submissions to one row per student per module-run (data.table, dplyr)	One analysis-ready table of 32,593 rows
Descriptive analysis	Outcome shares by group, medians, weekly activity, withdrawal-timing distribution	Show who enrols, how they engage and when they leave
Statistical testing	Chi-square test of independence with Cramér's V effect size	Rank which characteristics are most strongly linked to outcomes
Fair comparisons	Same modules across years; only students still registered on a due date; never-clickers kept separate	Avoid misleading comparisons
Segmentation	Four engagement segments, plus three student segments using each student's last active day	Locate non-completers, and find the silent stayers

Tools: R (tidyverse, data.table, scales, knitr) in an R Markdown report. Exams were excluded from the assessment analysis because exam submissions are recorded only for modules CCC and DDD.

4. Findings

4.1 Overall outcomes

Final result of all 32,593 registrations



Method	Count and share of each final result across all registrations; withdrawal by year, overall and on the same modules only.
Output	Pass **37.9%** (12,361) · Distinction **9.3%** (3,024) · Fail **21.6%** (7,052) · Withdrawn **31.2%** (10,156). Completed **47.2%**. Withdrawal **26.8%** in 2013J vs **34.0%** in 2014J; on the six modules that ran in both years, **26.8%** → **31.4%**. It rose in 6 of 7 modules.
What it means	Fewer than half of registrations succeed and withdrawal is rising across the institution. About a third of the headline rise is due to CCC joining in 2014.

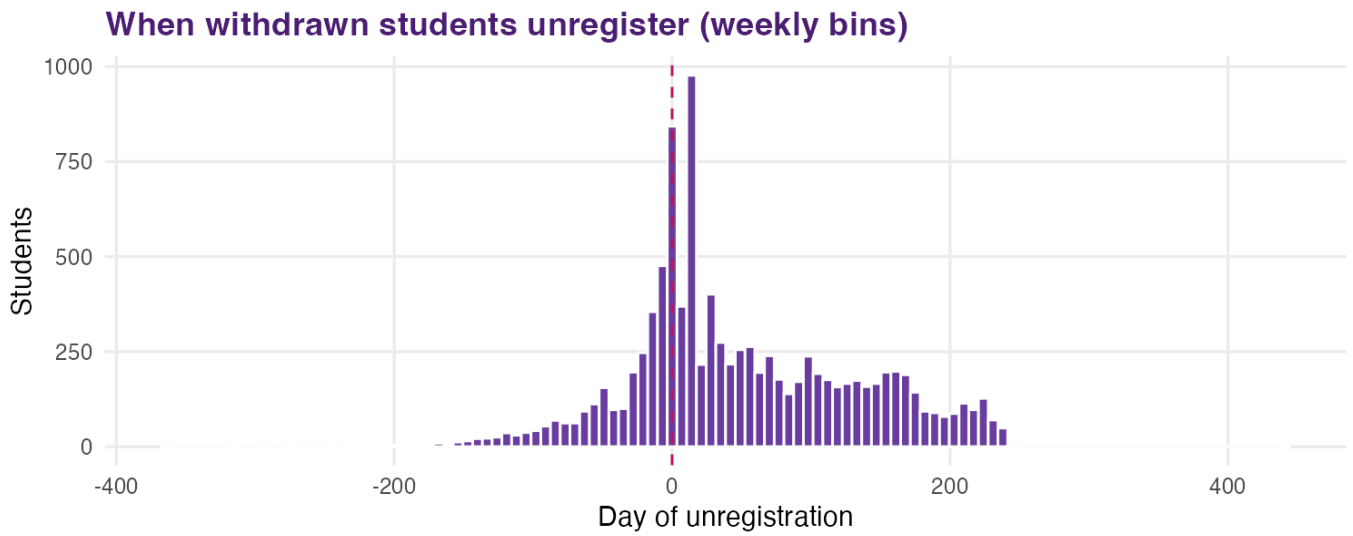
4.2 Who enrols and what matters most

Method	Profile of registrations; chi-square test of each characteristic against final result, with Cramér's V (0 = no link, 1 = perfect).
Output	

Characteristic	Cramér's V	Example
Credit load	0.127	Withdrawal 25.4% (≤60) → 38.7% (61–120) → 51.3% (>120)
Repeat attempts	0.123	Fail 31.7% for repeats vs 20.2% first attempts
Module	0.121	Withdrawal 11.5% (GGG) to 44.5% (CCC)
Highest education	0.102	Withdrawal 42.9% (no formal quals) vs 23.6% (postgraduate)
Deprivation band	0.094	Withdrawal 37.2% (0–10%) vs 25.9% (90–100%)
Gender	0.023	Almost no link

What it means	All links are significant ($p < 0.001$) but **all are weak** (V below 0.13): who a student is explains little on its own. The top three are practically tied, and two of them, workload and module, are in the university's control. Gender barely matters.

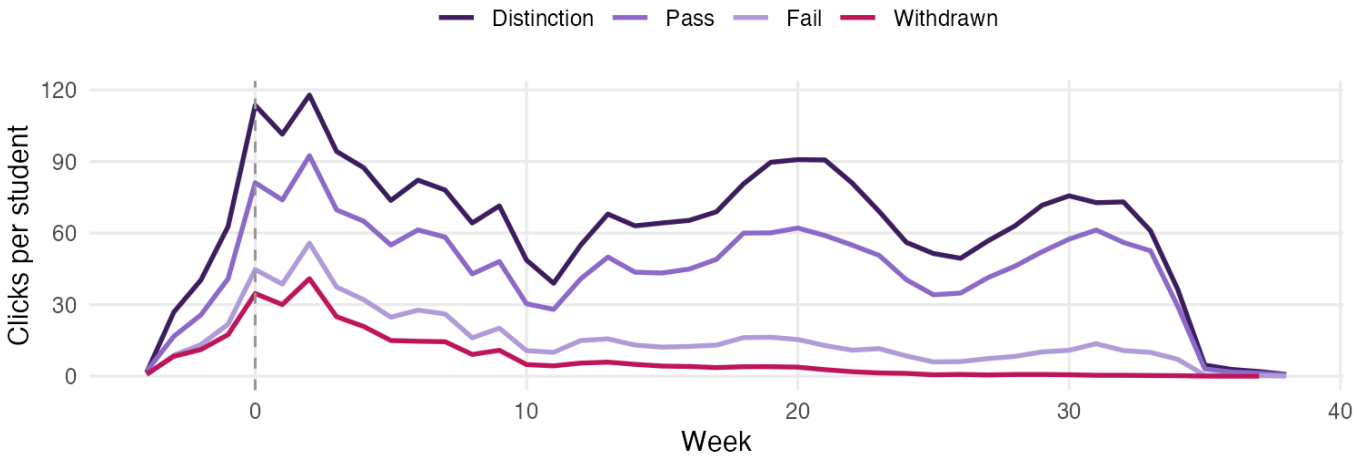
4.3 When students withdraw



Method	Distribution of unregistration day for the 10,063 withdrawn registrations with a recorded date.
Output	**30.7%** withdraw on or before day 0 · **50.9%** by day 30 · **61.9%** by day 60 · a smaller second wave of **14.9%** between days 121 and 180. In BBB, **41.5%** of withdrawals happen on or before day 0.
What it means	Withdrawal is front-loaded. Support that starts mid-module arrives too late for over half of those who leave.

4.4 How students engage

Average weekly clicks per student, by final result



Method	Click log summarised per student; weekly clicks per student by outcome; fixed bands of clicks in days 0–13.
Output	Median clicks: Distinction **1,896**, Pass **1,343**, Fail **317**, Withdrawn **89**. **3,365 (10.3%)** never clicked; **88.8%** of them withdrew. Zero clicks in days 0–13: **67.8%** withdrew vs **16.5%** with 150+. Clicking before day 0 barely matters once never-clickers are set apart (**24.2%** vs **25.9%** withdrawal). Pass students made **7.5×** more oucontent clicks than withdrawn students.
What it means	Healthy and troubled students are distinguishable from the first two weeks. The real warning sign is not engaging at all; troubled students mainly use the platform far less.

4.5 Assessment behaviour

Share submitting each of the first four assessments



Method	First-assessment submission and timing, counting only students still registered on its due date; submission across the first four non-exam assessments.
Output	7,208 (22.1%) have no first submission, but **4,405** of them had already left before it was due. Among students still registered: missed → **43.1%** withdrew, **4.7%** completed; late → 27.1% / 46.7%; on time → **13.4%** / **65.6%**. Fail group submission: 79.2% → 63.0% → 52.0% → 42.1%; Pass stays above 95%.
What it means	A missed first assessment is a strong, fair warning sign, and failing students fade away one assessment at a time.

4.6 What "Fail" actually looks like

Method	Medians of submission rate and average score by outcome; share of submissions made after the due date (banked results excluded).
Output	Median submission rate: Fail **36.4%** vs Pass **100%**. Median score: Fail **66.2** vs Pass **78.2**. **56.4%** of failing students submitted fewer than half. Late submissions: Fail **39.3%** vs Pass **26.8%**.
What it means	The gap is in whether students submit, not how well they score. Most failures are students who stopped taking part.

4.7 Where the problem concentrates

Method	Four behaviour-based engagement segments.
Output	

Segment	Students	Completion	Share of non-completers
1. Never clicked	3,365	0.1%	19.5%
2. Clicked, missed first assessment	3,890	3.4%	21.8%
3. Submitted first, then fewer than half	5,917	0.7%	34.2%
4. Submitted at least half	19,421	78.3%	24.5%

What it means	**Two universities in one:** segments 1–3 are 40.4% of students but 75.5% of non-completers, and only 1.3% of them finish; segment 4 (60%) finishes at 78.3%. Withdrawing also stops students clicking and submitting, so this shows where losses sit, not what caused them.

4.8 Deprivation and workload together

Method	Completion by credit load × deprivation group; deprivation gap within engagement segments.
Output	0–30% band with >120 credits: **22.4%** complete (821 students) vs **58.3%** for the 70–100% band with ≤60 credits. Overall gap 0–30% vs 70–100%: **15.9 points** (38.3% vs 54.2%), narrowing to **8.3 points** (73.4% vs 81.7%) among students who submitted at least half.
What it means	Disadvantage and overload compound. Much of the deprivation gap is a gap in engagement, which the university can act on.

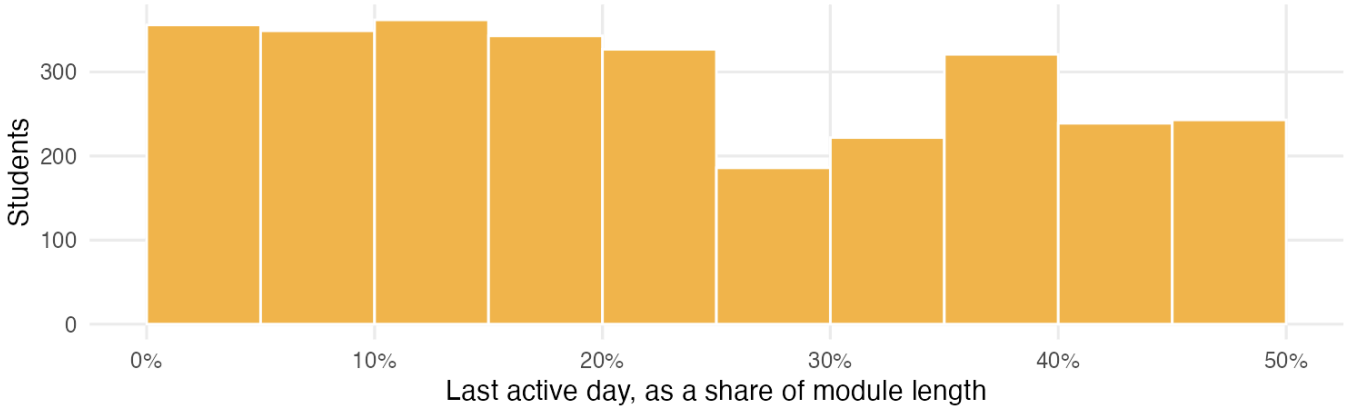
4.9 A third group: the silent stayers **OUR INNOVATION**

Three segments of students



Method	Official records only show "withdrew" or "stayed". Using each student's last click day as a share of module length, we separate students who did not withdraw into **active stayers** (still active after halfway) and **silent stayers** (no clicks after halfway, or never clicked), and profile them as their own segment.
Output	<table><tr><th>Segment</th><th>Share</th><th>Submitted first</th><th>Avg score</th><th>>120 credits</th><th>Repeat</th><th>Below A level</th><th>Most deprived</th></tr><tr><td>**Active stayers**</td><td>58.6%</td><td>98.4%</td><td>76.1</td><td>4.9%</td><td>10.7%</td><td>36.4%</td><td>27.9%</td></tr><tr><td>**Silent stayers**</td><td>**10.2%**</td><td>**60.9%**</td><td>**64.8**</td><td>**5.1%**</td><td>**19.7%**</td><td>**53.7%**</td><td>**40.7%**</td></tr><tr><td>**Official quitters**</td><td>31.2%</td><td>44.8%</td><td>64.1</td><td>11.4%</td><td>14.5%</td><td>47.0%</td><td>37.4%</td></tr></table> 99.8% of silent stayers (3,317 of 3,325) end with a Fail; they are 47% of all fails. Silent stayers range from 2.8% of students in AAA to 15.5% in GGG.
What it means	Silent stayers are not quitters: they **start well** (most hand in the first assignment, with decent marks), then **fade in the first half** of the module (median last click on day 53, about week 8). **Workload is not their problem**; they are more often repeat, less-qualified and deprived students. Because they are still registered, the university can still reach them, so they need their own strategy.

When silent stayers made their last click



	Official quitters	Silent stayers
When they are lost	Before the start and in the first month	At any point in the first half; typically around week 8
What stands out	Never engage; heavy workload	Started well, then faded; more repeat, less-qualified and deprived students
Strategy	**Fix the front door**	**Re-engage in the middle**
Actions	Personal call to students with no clicks by day 14; workload check at sign-up	Tutor contacts any student silent for 3 weeks; catch-up plan or deadline extension; option to pause and continue in the next presentation instead of failing; study-skills and peer support

The data shows what these students did, not why. The profile suggests confidence, study skills or life events matter more than workload for this group, but the university would need to ask them to be sure.

5. Our solution: where to focus in 2016

The data shows where the problem is. It cannot show which specific actions will work, so each focus area comes with a measure from this same data to track whether it helps.

Focus area	Why (evidence)	What to do	How to track it
1. The front door (to day 30)	50.9% of withdrawals by day 30; 10.3% never click and 88.8% of them withdraw; no clicks in days 0–13 → 67.8% withdraw	A person calls every student with no clicks by day 14	Share of students with no clicks in days 0–13
2. First assessment checkpoint	Still-registered students who miss it: 4.7% complete, vs 65.6% on time	Follow up immediately with every student who misses or is late on the first assessment	First-assessment submission and on-time rate
3. Silent stayers **NEW**	10.2% of students; 99.8% fail; started well (61% submitted first, avg score 65), then went quiet	Tutor contact after 3 silent weeks; catch-up plan; pause-and-resume instead of Fail; study-skills and peer support	Silent-stayer rate per module-run
4. Workload at sign-up	Withdrawal 51.3% above 120 credits vs 25.4% at ≤60; 22.4% completion for 0–30% band at >120	A workload "speed limit": a conversation before anyone signs up for more than 120 credits	Completion by credit group
5. Fix courses, not students	CCC 44.5% withdrawal and 37.8% completion vs AAA 71.0%; GGG has the most silent stayers (15.5%); BBB 41.5% of withdrawals before day 0	Rebuild the start of CCC and DDD using what works in AAA; look at why GGG students go silent	Withdrawal and silent-stayer rate per module-run
6. Change the scoreboard	Official figures hide silent stayers; exam results only for CCC and DDD; 1,111 missing IMD bands; no withdrawal reasons	Report students active on day 30 and silent stayers each year; record exams, deprivation and withdrawal reasons	Completeness of these figures

What kind of institution are we?

Two universities in one. An engaged university, where 60% of students keep taking part and 78% of them finish, sits beside a disengaged one, where 40% never start properly or fade away and only 1.3% finish. Most students who quit are lost at the front door, and a hidden group of silent stayers fades mid-course while still on the books. The completion problem is **early, visible and concentrated**: completion is won or lost in the first weeks of a module.

6. Limitations and assumptions

- This is a descriptive analysis: it shows associations, not proof of cause. It is not a forecast or an early-warning model, as the Board asked.
- Withdrawing stops students clicking and submitting, so engagement segments partly reflect withdrawal. Fair checks count only students still registered at the time.
- "No clicks after halfway" is a simple cut-off for silent stayers; a student could still study offline.
- 1,909 banked assessment results count as submitted but are left out of late/on-time checks.
- Exam submissions are recorded only for CCC and DDD, so exams were excluded from the assessment analysis.

- Page types (e.g. oucontent) and assessment types (TMA, CMA) are not defined in the data and are reported by label only.
- Assumed: day 0 is the module start, and the 0–10% band is the most deprived (it has the lowest completion); neither is stated in the data.
- 93 withdrawn registrations have no unregistration date and are excluded from the timing analysis.
- Figures count registrations, not unique people; 3,538 students appear in more than one module-run.
- The data covers 22 module-runs across 2013 and 2014 only. A day-14 "traffic light" idea is in the analysis appendix, outside this brief.